MATH 2431 | Honors Probability | Spring 2026
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Problem Set 7

Submission:
Friday, 17/04/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.
Only problems marked with $(*)$ will be graded.

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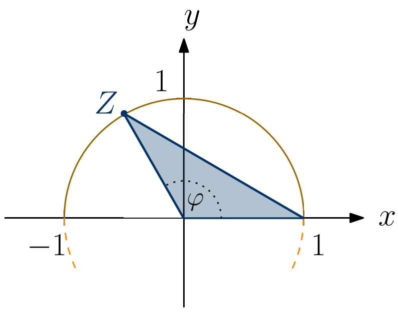
1. **Expectation of transformed random variables**

   (a) 1 $ is put on the table. We toss a coin that shows "heads" with probability $p \in (0,1)$. Every time the coin shows "heads", the amount of money on the table is doubled, and we stop the game after n steps. Let X be the amount of money on the table after the experiment is over. Calculate $\mathbf{E}[X]$ and $\mathrm{Var}[X]$.

   (b) Consider a circle centered at $(0,0)$ with radius 1. A point Z is chosen uniformly on the upper half of the circle. Let A denote the area of the triangle with coordinates $(0,0)$, $(1,0)$ and Z (see the sketch below). Calculate $\mathbf{E}[A]$ and $\mathrm{Var}[A]$.



   Hint: Represent Z in polar coordinates, $Z = (\cos(\Phi), \sin(\Phi))$, where $\Phi \sim \mathscr{U}([0,\pi])$. Use the identity $\cos(2\varphi) = 1 - 2\sin^2(\varphi)$ for $\varphi \in \mathbb{R}$.

2. **Markov and Čebyšev inequalities**

   (a) A biased coin lands on heads with probability $\frac{1}{3}$. This coin is tossed 200 times. Use Markov's inequality to give an upper bound on the probability that the coin shows heads at least 120 times. Improve this bound using Čebyšev's inequality.

   (b) Find a random variable $X : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ and a value $a > 0$ such that Čebyšev's inequality is tight, i.e.:
$$\mathbf{P}\big[|X - \mathbf{E}[X]| \geq a\big] = \frac{\mathrm{Var}[X]}{a^2}.$$
   Hint: It suffices to determine the law $\mathbf{P}_X$. Look for a discrete distribution that assigns positive probability to only three distinct values.

3. $(*)$ **Chernoff bounds** [4 Points]

   Let $(\Omega, \mathscr{F}, \mathbf{P})$ be a probability space and $X : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ a random variable such that the moment generating function $\psi_X(t)$, defined by
$$\psi_X(t) = \mathbf{E}\left[e^{tX}\right],$$
   exists for every $t \in \mathbb{R}$.

(a) Let $a \in \mathbb{R}$. Show that for $t > 0$, we have

$$\mathbf{P}[X \geq a] \leq e^{-ta}\psi_X(t),$$

whereas for $t < 0$, we have

$$\mathbf{P}[X \leq a] \leq e^{-ta}\psi_X(t).$$

Hint: The function $\exp : \mathbb{R} \to \mathbb{R}$ is strictly increasing. Use Markov's inequality.

(b) Calculate $\psi_X(t)$ for $X \sim \mathcal{N}(0,1)$.

(c) Show that for $X \sim \mathcal{N}(0,1)$, we have

$$\mathbf{P}[X \geq a] \leq e^{-\frac{a^2}{2}}, \text{ for } a > 0,$$

and

$$\mathbf{P}[X \leq a] \leq e^{-\frac{a^2}{2}}, \text{ for } a < 0.$$

4. $(*)$ **Joint distributions** [4 Points]

(a) For the examples i. and ii. below, decide (with a mathematical explanation) whether the function $F : \mathbb{R}^2 \to \mathbb{R}$ can be the joint cumulative distribution function of a pair (X, Y) of random variables. For the case where F is indeed a joint cumulative distribution function, determine the cumulative distribution functions of X and Y.

i. $F(x,y) = \begin{cases} 1 - e^{-x-y}, & x, y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$

ii. $F(x,y) = \begin{cases} 1 - e^{-x} - xe^{-y}, & 0 \leq x \leq y \\ 1 - e^{-y} - ye^{-y}, & 0 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$

(b) Let X, Y be discrete random variables with joint probability mass function

$$p_{X,Y}(j,k) = \mathbf{P}[X = j, Y = k] = \begin{cases} C \cdot \left(\frac{1}{2}\right)^k, & \text{for } k = 2, 3, ..., j = 1, 2, ..., k-1, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the constant C and the marginal probability mass functions p_X and p_Y.